Audibility of group delay distortions

J. Blauert and P. Laws

Citation: The Journal of the Acoustical Society of America **59**, S70 (1976); doi: 10.1121/1.2002850
View online: https://doi.org/10.1121/1.2002850
View Table of Contents: https://asa.scitation.org/toc/jas/59/S1
Published by the Acoustical Society of America

is a powerful adaptive technique designed for problems of this type and has been used as part of an ongoing study to develop a practical protocol for the prescriptive fitting of a wearable master hearing aid. Preliminary trials with five subjects showed superior performance for the fitted wearable master hearing aid than that obtained with the subjects' own aid under comparable conditions of speech in noise. [Research supported by PHS Contract No. NIH-N01-NS-4-2323.]

4:45

FF12. **Audibility of group delay distortions.** J. Blauert (Ruhr-Universität, 4630 Bochum, Germany F.R.) and P. Laws (Gesamthochschule, 4100 Duisburg, Germany F.R.)

We report upon exemplary measurements of group-delay distortions as induced into electroacoustical transmission by the fact that earphones and loudspeakers are not necessarily minimum phase systems. Additional psychoacoustical tests showed that the measured distortions are on the order of the limit of perceptibility or exceed this limit. The limit of perceptibility of group-delay distortions depends strongly upon the kind of signal, the shape of the group-delay curve, and the state of training of the subjects. For short sound pulses and group-delay distortions as produced by second-order all-pass filters we found in diotic presentation, with one all-pass at frequencies between 1 and 4 kHz, a limit of group-delay perceptibility between 1 and 2 msec for untrained subjects and 0.4 msec for trained subjects. Further experiments with up to 20 all-pass filters showed that for group-delay curves with multiple maxima and minima the limit of perceptibility has a value of about 0.6 msec (peak to valley) for trained subjects.

5:00

FF13. **Speech audiometry via bone conduction.** B. Edgerton, J. L. Danhauer, and T. Barnard (Department of Speech Communication, Bowling Green State University, Bowling Green, OH 43403) and R. Beattie (Department of Speech, Long Beach State, Long Beach, CA 90808)

This study was designed to determine (1) values for SRT's obtained via bone conduction using the NU cassette recording, (2) the relationship between BC PTA's and BC SRT's for normals, (3) short-term reliability of BC SRT's, and (4) characteristics of articulation functions for spondees obtained by BC and by AC. An average BC SRT was determined for 50 normally hearing young adults using mastoid placement. While relationships between BC PTA's and BC SRT's were not in agreement with those indicated by earlier investigators, the average SPL values for the BC SRT's were. Test-retest reliability for BC SRT's was very high. A comparison of articulation functions indicates that the perception of speech as a function of intensity does not differ significantly for air or bone conducted transmission systems.

THURSDAY, 8 APRIL 1976 FEDERAL ROOM, 2:00 P.M.

Session GG. Speech Communication VI: Speech Analysis

Thomas H. Crystal, Chairman

Institute for Defense Analyses, Princeton, New Jersey 08540

Invited Paper

2:00

GG1. **Vocal-tract determinants of resonance frequencies and bandwidths.** G. Fant [Department of Speech Communication, Royal Institute of Technology (KTH), S-100 44 Stockholm 70, Sweden]

This is a review of recent years' development of the theory of vocal transmission focusing on the spatial distribution of energy, resonance bandwidths, perturbations in area and in length dimensions with applications to nonuniform scaling of vocal-tract dimensions, and discussions of "resonance-cavity" relations. In addition to the well established influence of the vocal-wall impedance on the tuning of very low F_1 resonances, there are also indications that F_1 of [a]-type vowels could be significantly influenced by the tuning of the vocal-wall mass shunt which, according to our measurements, should be heavier in the region of the lower pharynx than anywhere else. New measurements of the closed-tract resonance and bandwidth for males and females are reported and also some observed differences in male–female cavity size and shape.

Contributed Papers

2:24

GG2. **Uniqueness of articulations determined from acoustic data.** Lloyd Rice and Peter Ladefoged (Phonetics Laboratory, UCLA, Los Angeles, CA 90024)

What is the minimum set of acoustic data that determines a unique shape of the vocal tract? This depends in part on the degree of accuracy with which the vocal tract shape is to be specified (the number of points on the surface of the vocal tract that are determined), in part on the kind of acoustic data (whether both frequencies and bandwidths of the formants are known), and in part on the constraints that are imposed on the possible shapes of the vocal tract, so that only potentially observable shapes are considered. We assume that the degree of accuracy required involves specifying 18 points along the vocal tract, and that the articulatory constraints are those that have been observed in non-nasalized vowels. We then show the extent to which we can determine the shape of the vocal tract, given only two or three formant frequencies and no bandwidth information.